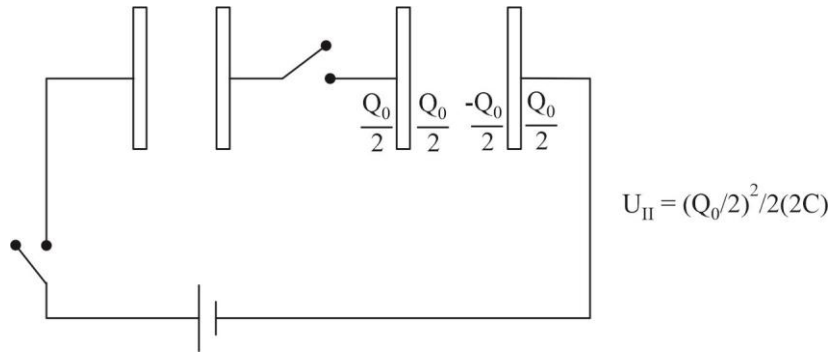


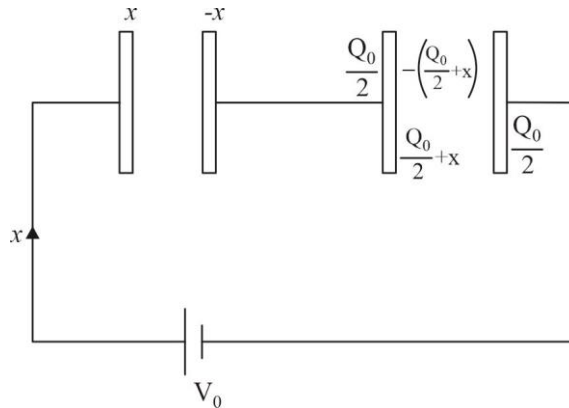
Solutions to JEE Advanced - 5 | JEE 2022 | Paper - 2

PHYSICS

1.(BCD) Before



After



$$\Rightarrow V_0 = \frac{x}{C} + \frac{\frac{Q_0}{2} + x}{2C} \Rightarrow \frac{3x}{2C} + \frac{CV_0}{4C} = V_0$$

$$\Rightarrow \frac{3x}{2C} = V_0 - \frac{V_0}{4} = \frac{3V_0}{4}$$

$$x = \frac{CV_0}{2}$$

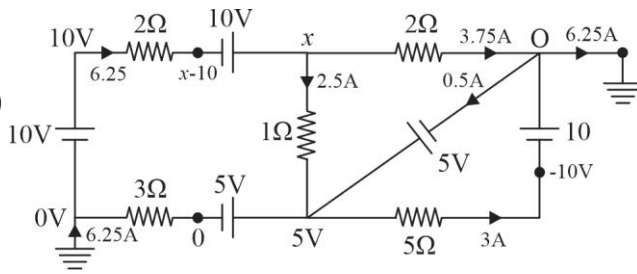
$$\Rightarrow V_I = \frac{x}{C} \quad V_{II} = \frac{\frac{Q_0}{2} + x}{2C} = \frac{V_0}{2} = \frac{V_0}{2}$$

$$\Rightarrow U_{II(\text{final})} - U_{II(\text{initial})} = \frac{(CV_0)^2}{4C} - \frac{(CV_0/2)^2}{4C} = \frac{3C^2V_0^2}{4(4C)} = \frac{3C^2V_0^2}{16C}$$

2.(BC) B along z-axis : L_{eff} and B are parallel $F = 0$ B along y-axis : $L_{\text{eff}} = 2L$

$$F = IB(2L)$$

B along x-axis : $F = IB(2L)$

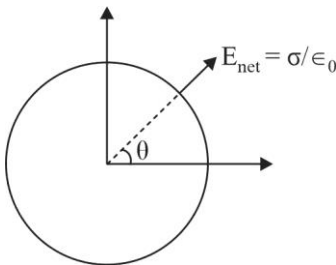


$$\frac{10-(x-10)}{2} = \frac{x-0}{2} + \frac{x-5}{1} \Rightarrow x = 7.5V$$

$$i_{5\Omega} = \frac{15}{5} = 3A \quad i_{2\Omega} = \frac{7.5}{2} = 3.75$$

$$i_{3\Omega} = 0$$

4.(CD)



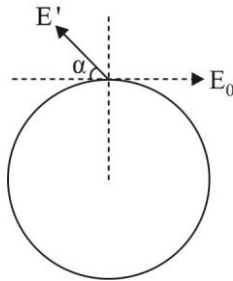
$\Rightarrow$ Electric field due to charges other than dA at P is $\frac{\sigma}{2\epsilon_0}$

$$\text{So, force} = \frac{\sigma}{2\epsilon_0} \times \sigma dA$$

$$E' \sin \alpha = \frac{\sigma}{\epsilon_0}$$

$$E' \cos \alpha = E_0$$

$$E' = \sqrt{E_0^2 + \frac{\sigma^2}{\epsilon_0^2}}$$



5.(ABD) $i = \frac{E}{R}(1 - e^{-Rt/L})$

$$\Rightarrow t = t_0 \Rightarrow \frac{1}{2} = 1 - e^{-Rt/L} \Rightarrow t_0 = \frac{L}{R} \ln(2)$$

$\Rightarrow$ Rate at which energy transferred to inductor is maximum when $V_L = V_R \Rightarrow t_0 = t$

$$\frac{1}{2} \cdot \frac{1}{2} L \left(\frac{E}{R} \right)^2 = \frac{1}{2} L \left(\frac{E}{R} \right)^2 \left(1 - e^{-Rt/L} \right)^2$$

$$\Rightarrow 1 - e^{-Rt/L} = \frac{1}{\sqrt{2}} \Rightarrow e^{-Rt/L} = 1 - \frac{1}{\sqrt{2}} = \frac{2 - \sqrt{2}}{2}$$

$$t = \frac{L}{R} \ln \left(\frac{2}{2 - \sqrt{2}} \right)$$

$$t = t'_0 = \frac{L}{R} [\ln(2) - \ln(2 - \sqrt{2})] \Rightarrow t_0 - t'_0 = \frac{L}{R} \ln(2 - \sqrt{2})$$

$$q = \int_0^{t_0} i dt = \frac{E}{R} \times \frac{L}{R} \ln(2) - \frac{EL}{R^2} (1 - e^{-\ln 2}) = \frac{EL}{R^2} \ln(2) - \frac{EL}{2R^2} = \frac{EL}{2R^2} (2\ln(2) - 1)$$

$$6.(ABD) \frac{1}{z} = \sqrt{\frac{1}{R^2} + \left(\omega C - \frac{1}{\omega L}\right)^2}$$

Resonance

$$\omega = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{10^{-9}}} = 3.16 \times 10^4 \frac{\text{rad}}{\text{s}}$$

$$I = \frac{V}{R} = \frac{31.6}{10} = 3.16 \text{ A}$$

$$I_L = \frac{V}{\omega L} = \frac{31.6}{31.6} = 1 \text{ A}$$

$$I_C = V\omega C = 1 \text{ A}$$

7.(C)

8.(D)

Total flux $\phi = B_0(1 - \alpha z)\pi r_0^2 + LI = \text{constant}$ initially $z = 0$ & $I = 0$

$$\Rightarrow \phi = B_0\pi r_0^2 \Rightarrow I = \frac{B_0\alpha\pi r_0^2}{L} z$$

Lorentz force acting on the ring is

$$F_z = -B_r I 2\pi r_0 = \frac{-2B_0^2\pi^2 r_0^4 \alpha \beta z}{L}$$

$$\text{Mean position} \Rightarrow z_0 = \frac{mgL}{2B_0^2\pi^2 r_0^4 \alpha \beta}$$

Amplitude = z_0

9.(D)

10.(A)

$$B = \frac{2\mu_0 n I R^2}{2\left(R^2 + \left(\frac{R}{2}\right)^2\right)^{3/2}} = \frac{\mu_0 n I}{R\left(\frac{5}{4}\right)^{3/2}} = \left(\frac{4}{5}\right)^{3/2} \frac{\mu_0 n I}{R}$$

11.(C)

12.(B)

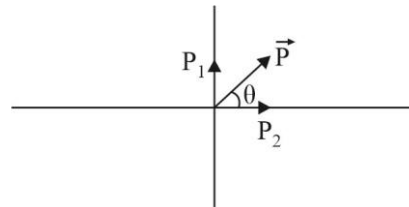
$$P_1 = q(2l)$$

$$P_2 = ql$$

$$P = \sqrt{5}ql$$

$$\tan \theta = \frac{P_1}{P_2} = 2$$

ω is maximum when potential energy is minimum



$$-\sqrt{5}qlE_0 \cos \theta = \frac{1}{2}I\omega^2 - (\sqrt{5}ql)E_0$$

$$\frac{1}{2}I\omega^2 = \sqrt{5}qlE_0 \left(1 - \frac{1}{\sqrt{5}}\right) \Rightarrow \omega^2 = 2qlE_0(\sqrt{5}-1)/I = \frac{24qlE_0(\sqrt{5}-1)}{5ml^2}$$

1.(1.5) After 1st contact

$$q/C_1 = (Q-q)/C_2 \Rightarrow \frac{0.6}{C_1} = \frac{0.4}{C_2} \Rightarrow 6C_2 = 4C_1$$

C_1 : Capacitance of conductor

C_2 : Capacitance of plate

$$q = \frac{QC_1}{C_1 + C_2}$$

After 2nd contact

$$\frac{q'}{C_1} = \frac{(Q+q-q')}{C_2}$$

$$q'C_2 = C_1Q + C_1q - C_1q' \Rightarrow q' = \frac{C_1Q + C_1q}{C_1 + C_2}$$

After 3rd contact

$$q'' = \frac{C_1Q}{C_1 + C_2} + \left(\frac{C_1}{C_1 + C_2}\right)^2 Q + \left(\frac{C_1}{C_1 + C_2}\right)^3 Q$$

$$\text{After } \infty \text{ steps } \Rightarrow q = \frac{QC_1}{C_1 + C_2} \left(\frac{C_1 + C_2}{C_2}\right) = Q \frac{C_1}{C_2} = 1.5Q$$

$$2.(400) F_2 = \int E_1 dq_2 = \int E_1(\sigma_2 dA_2) \cos \theta$$

$$= \sigma_2 \int E_1 dA_2 \cos \theta = \sigma_2 \phi \Rightarrow \phi = \frac{24}{6 \times 10^{-2}} = 400 \frac{Nm^2}{C}$$

$$3.(1) \Delta q = \frac{\Delta \phi}{3R}; l = \frac{1}{2}a_0 \times \frac{\sqrt{3}a}{2a_0} = \frac{\sqrt{3}a}{4}$$

$$\Delta q = \frac{B_0 \times \frac{1}{2} \times \frac{a}{2} \times \frac{\sqrt{3}a}{4}}{3R} = \frac{\sqrt{3}a^2 B_0}{48R}$$

$$= \frac{\sqrt{3} \times 16 \times 5}{48 \times \frac{5}{\sqrt{3}}} = 1 \text{ coulomb}$$

$$4.(30) \quad dB = \frac{\mu_0}{2} \frac{di y^2}{(y^2 + x^2)^{3/2}} \quad (\text{due to differential element})$$

$$= \frac{\mu_0 \sigma \omega}{2} \frac{y^3}{(y^2 + x^2)^{3/2}} dy$$

$$B = \int_0^R dB = \frac{\mu_0 \sigma \omega}{2} \left[\frac{R^2 + 2x^2}{\sqrt{R^2 + x^2}} - 2x \right]$$

For centre; $\frac{\mu_0 \sigma \omega R}{2}$ (due to complete disc)

For annulate disc

$$B = \frac{\mu_0 \sigma \omega}{2} (R_2 - R_1)$$

$$= \frac{4 \times 5 \times 3}{2} = 30 \text{ Tesla}$$

$$5.(50) \quad i = 0 \Rightarrow Blv = \frac{q}{C} \quad \dots(i) \quad \text{when force on rod becomes zero}$$

$$-iBl = \frac{mdv}{dt} \quad \dots(ii)$$

$$(Q_0 - q)Bl = mv$$

$$\Rightarrow v = \frac{Q_0 Bl}{m + B^2 l^2 C} = \frac{20 \times 10^{-6} \times 30 \times 10^2 \times 1}{1 + 10^4 \times 10^{-5} \times 2}$$

$$= 5 \times 10^{-2} \text{ m/s}$$

$$= 50 \text{ mm/s}$$

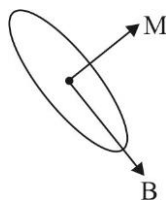
6.(4) Cons. Of energy

$$-\sqrt{34} \sqrt{136} \cos 90^\circ + 0$$

$$= -\sqrt{34} \sqrt{136} \cos 0^\circ + \frac{1}{2} \left(\frac{I}{2} \right) \omega^2$$

$$\Rightarrow \omega^2 = \frac{34 \times 2 \times 4}{7} = 16$$

$$\omega = 4 \text{ rad/s}$$

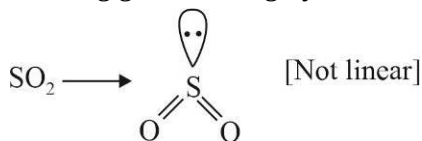


CHEMISTRY

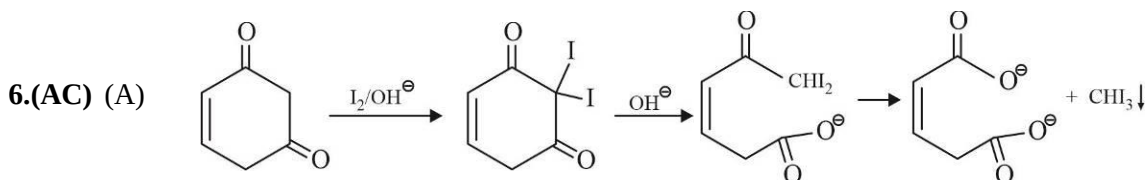
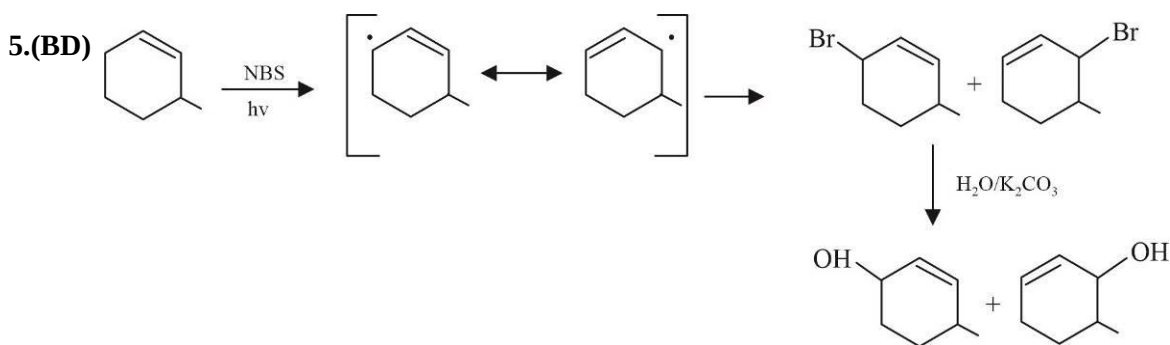
1.(ABD) Facts

2.(AB) Isotonic solutions have same osmotic pressure and same concentration (Molarity). As the solvent can be different in two isotonic solutions the molality can be different. Hence elevation in boiling point and depression in freezing point can be different as they depend on molality.

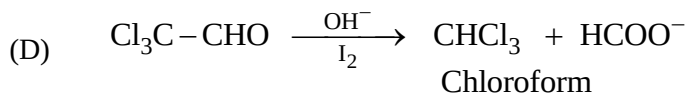
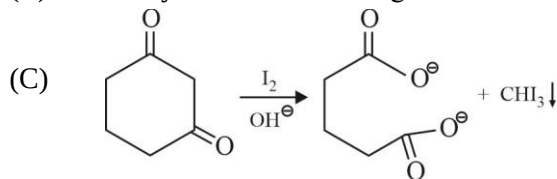
3.(AB) SO_2 is used as bleaching agent in moist conditions, disinfectant and preservative. It is colorless and a pungent smelling gas and is highly soluble in water (refer NCERT).



4.(A) Compounds I and II are 2° benzylic and 3° RX respectively hence will follow $\text{S}_{\text{N}}1$ mechanism. Compound III is achiral hence no optical inversion will take place.



(B) t-butyl alcohol doesn't give iodoform test



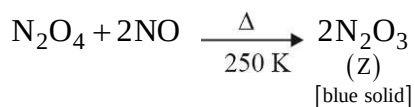
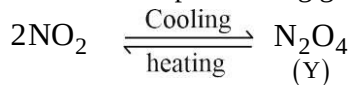
7.(B) Arginine is the most basic amino acid because it contains guanidine side group.

8.(B) On the complete acidic hydrolysis of the given polypeptide, the amino acids obtained are

1. Glycine
2. Serine
3. Methionine
4. Phenylalanine

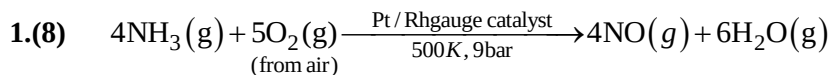
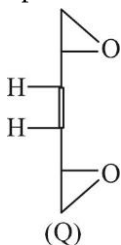
Out of the above four amino acids methionine and phenylalanine are essential.

9.(D) 10.(C)

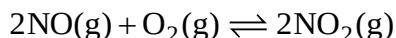
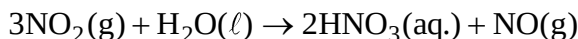
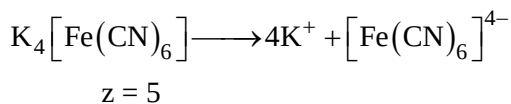
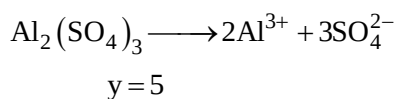
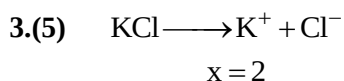
Lead Nitrate upon heating gives PbO, O₂ (A) and NO₂ (X) gas

11.(B) 'S' should be alkene. Also C2 and C5 should have opposite configuration for S to be meso.

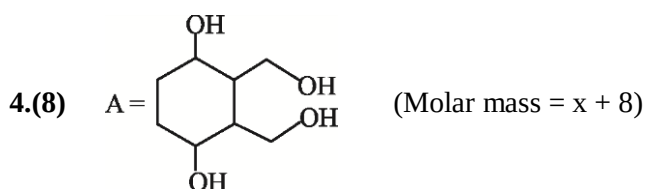
12.(D) In compound 'T' C2 and C5 should have opposite configuration similar to S to be meso it is possible with option D only



It is clear from the reaction that 12 moles of ammonia produces 12 moles of NO.

In next step 12 moles of NO will be oxidized to 12 moles of NO₂Finally 12 moles of NO₂ produced is dissolved in water to produce 8 moles of HNO₃.2.(4) Body diagonal length is $\sqrt{3}a$ while the nearest distance between octahedral and tetrahedral voids is $\frac{\sqrt{3}a}{4}$.

$$\text{So, } \frac{y+z}{x} = \frac{5+5}{2} = 5$$



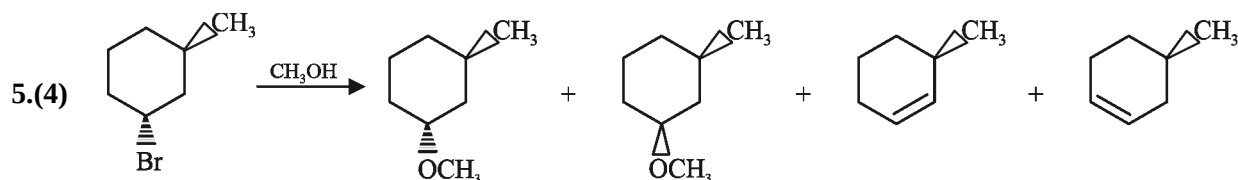
No. of alcohol = No. of moles of Ac_2O consumed = 4

Change in molar mass per OH group = 42

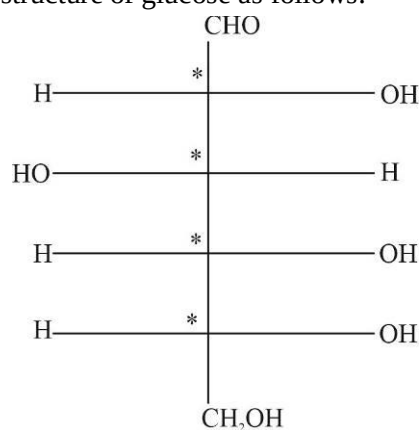
Molar mass of B = $x + 8 + (42 \times 4) = x + 176$

$$y = x + 176$$

$$\frac{y - x}{22} = 8$$

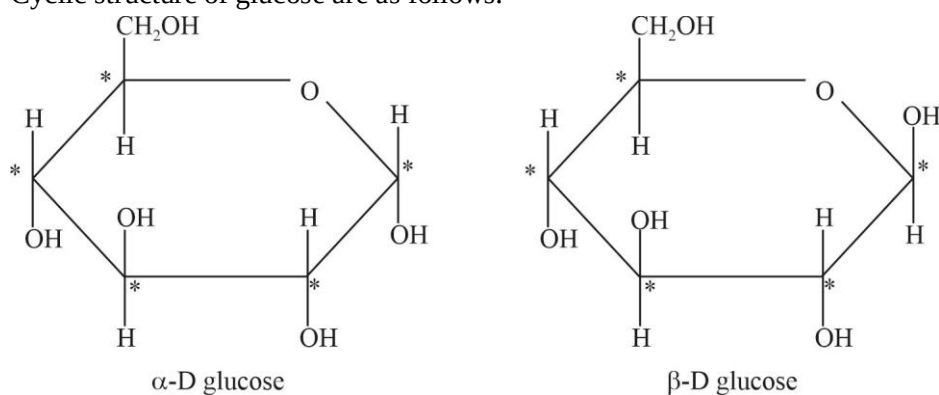


6.(9) Linear structure of glucose as follows:



Number of chiral carbons (C^*) = 4.

Cyclic structure of glucose are as follows:-



Number of chiral carbons (C^*) in each isomer = 5

Therefore, value of z is 14.

MATHEMATICS

1.(ABCD) $f(x^2 + y) = (f(x))^2 + f(y)$

$$f(0) = (f(0))^2 + f(0) \quad f(0) = 0$$

for $y = 0$: $f(x^2) = (f(x))^2 \quad \dots(i)$

for $y = -x^2$: $0 = f(0) = (f(x))^2 + f(-x^2)$

$$\Rightarrow (f(x))^2 = -f(-x^2) \quad \dots(ii)$$

From (i) and (ii)

$$f(-x^2) = -f(x^2)$$

$$\Rightarrow f(-x) = -f(x) \quad \dots(iii)$$

Thus $f(x)$ is an odd function if $f(x)$ even also, then $f(-x) = f(x) \quad \dots(iv)$

$$\therefore f(x) = 0 \text{ for all } x \quad \{\text{by (iii) and (iv)}\}$$

Since $f(x)$ is continuous at $x = 0$,

$$\therefore \lim_{h \rightarrow 0} f(h) = 0 \Rightarrow \lim_{h^2 \rightarrow 0} f(x + h^2) = \lim_{h^2 \rightarrow 0} \{(f(h))^2 + f(x)\} = f(x)$$

$\Rightarrow$ it is continuous everywhere

Since $\lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{f(h)}{h}$ exists

$$\therefore \lim_{h \rightarrow 0} \frac{f(x+h^2) - f(x)}{h^2} = \lim_{h \rightarrow 0} \frac{(f(h))^2 + f(x) - f(x)}{h^2} = (f'(0))^2$$

$$\therefore f'(x) = (f'(0))^2 \Rightarrow f'(0) = (f'(0))^2$$

$$\Rightarrow f'(0) = 0 \text{ or } f'(0) = 1$$

$$\therefore f(x) = (f'(0))^2 x$$

Now (i) if $f'(0) = 0$, then $f(x) = 0 = x f'(0)$

(ii) if $f'(0) = 1$, then $f(x) = x = x f'(0)$

2.(BC) $\left(2^{\frac{\pi}{\cos^{-1}x}}\right)^2 - \left(a + \frac{1}{2}\right)2^{\frac{\pi}{\cos^{-1}x}} - a^2 = 0$

let $2^{\frac{\pi}{\cos^{-1}x}} = t \Rightarrow t \geq 2$

Now equation $t^2 - \left(a + \frac{1}{2}\right)t - a^2 = 0$ has one root ≥ 2 or greater than 2 and other root less than 2

$$\Rightarrow f(2) \leq 0$$

$$\Rightarrow 4 - \left(a + \frac{1}{2}\right)2 - a^2 \leq 0$$

$$4 - 2a - 1 - a^2 \leq 0$$

$$a^2 + 2a - 3 \geq 0$$

$$(a+3)(a-1) \geq 0 \Rightarrow a \leq -3 \text{ or } a \geq 1$$

3.(CD) (A) Consider $g(x) = \sin^{-1} f(x) - x$

$$\text{Since } g(0) = 0, g\left(\frac{\pi}{2}\right) = 0$$

$$\therefore \text{ there is at least one value of } \alpha \in \left(0, \frac{\pi}{2}\right) \text{ such that } g'(\alpha) = \frac{f'(\alpha)}{\sqrt{1-(f(\alpha))^2}} - 1 = 0$$

$$\text{i.e., } f'(\alpha) = \sqrt{1-(f(\alpha))^2} \text{ for atleast one } \alpha \in (0, \pi/2) \therefore \text{ false}$$

(B) Consider $g(x) = f(x) - \frac{2x}{\pi}$

$$\text{Since } g(0) = 0, g\left(\frac{\pi}{2}\right) = 0$$

$$\therefore \text{ there is at least one value of } \alpha \in \left(0, \frac{\pi}{2}\right) \text{ such that } g'(\alpha) = f'(\alpha) - \frac{2}{\pi} = 0$$

$$\text{i.e., } f'(\alpha) = \frac{2}{\pi} \text{ for atleast one value of } \alpha \text{ but may not be for all } \alpha \in \left(0, \frac{\pi}{2}\right)$$

$$\therefore \text{ false}$$

(C) Consider $g(x) = (f(x))^2 - \frac{2x}{\pi}$

$$\text{Since } g(0) = 0, g\left(\frac{\pi}{2}\right) = 0$$

$$\therefore \text{ there is at least one value of } \alpha \in \left(0, \frac{\pi}{2}\right) \text{ such that } g'(\alpha) = 2f(\alpha)f'(\alpha) - \frac{2}{\pi} = 0$$

$$\therefore \text{ true}$$

(D) Consider $g(x) = f(x) - \frac{4x^2}{\pi^2}$

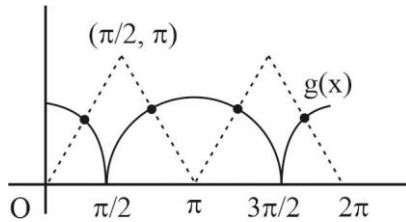
$$\text{Since } g(0) = 0, g\left(\frac{\pi}{2}\right) = 0$$

$$\therefore \text{ there is at least one value of } \alpha \in \left(0, \frac{\pi}{2}\right) \text{ such that } g'(\alpha) = f'(\alpha) - \frac{8\alpha}{\pi^2} = 0$$

$$\therefore f'(\alpha) = \frac{8\alpha}{\pi^2} \therefore \text{ true}$$

4.(ABCD) $f(x) = \cos^{-1}(\cos 2x)$, $g(x) = |\cos x|$

$f(x)$, and $g(x)$ both are even and periodic so $\max\{f(x), g(x)\}$ and $\min\{f(x), g(x)\}$ will also be periodic and even.



But $\max\{f(x), g(x)\}$ will be non-differentiable when $f(x) = g(x)$ no of points where $f(x) = g(x)$ are four in $[0, 2\pi]$

5.(BD) $f'(x) = 0 \Rightarrow 2\cos x + 2\cos 2x = 0$

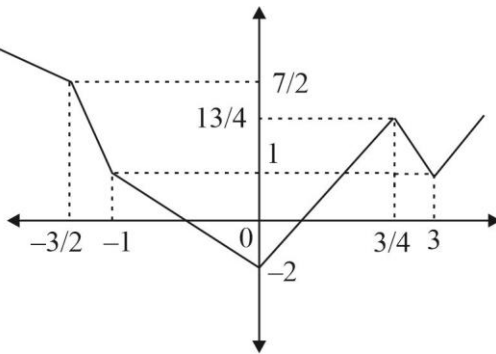
$\Rightarrow x = \frac{\pi}{3}, \pi$ are critical points

$f(x)$ is increasing in $\left[0, \frac{\pi}{3}\right]$ and decreasing in $\left[\frac{\pi}{3}, \frac{3\pi}{2}\right]$

Hence $f(0) = 0$, $f\left(\frac{\pi}{3}\right) = \frac{3\sqrt{3}}{2}$, $f(\pi) = 0$, $f\left(\frac{3\pi}{2}\right) = -2$

Least value $= -2$, Greatest value $= \frac{3\sqrt{3}}{2}$

6.(ACD)



7.(C) As $\det. (P - nI) = 0$

$\Rightarrow \begin{vmatrix} 1-n & 2 & 0 \\ 2 & 1-n & 0 \\ 0 & 0 & 1-n \end{vmatrix} = 0$

$\Rightarrow (1-n)(n-3)(n+1) = 0 \Rightarrow n = -1, 1, 3$

So $N = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{bmatrix} \Rightarrow Q = \begin{bmatrix} \frac{1}{3} & 0 & 0 \\ 0 & \frac{-1}{3} & 0 \\ 0 & 0 & 3 \end{bmatrix}$

8.(C) Given $P = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$$\text{So } P^2 = PP = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 5 & 4 & 0 \\ 4 & 5 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\text{Now } P^3 = P^2P = \begin{bmatrix} 5 & 4 & 0 \\ 4 & 5 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 13 & 14 & 0 \\ 14 & 13 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$P^4 = P^3P = \begin{bmatrix} 13 & 14 & 0 \\ 14 & 13 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 41 & 40 & 0 \\ 40 & 41 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\text{Also } P^5 = P^4P = \begin{bmatrix} 41 & 40 & 0 \\ 40 & 41 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 121 & 122 & 0 \\ 122 & 121 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\therefore \text{Tr.}(P) = 3, \text{Tr.}(P^3) = 27 = 3^3$$

$$\text{Tr.}(P^5) = 243 = 3^5 \Rightarrow \text{Tr.}(P^{2011}) = 3^{2011}$$

$$\text{Also } \text{Tr.}(P^2) = 11 = 3^2 + 2$$

$$\text{Tr.}(P^4) = 3^4 + 2$$

$$\text{Tr.}(P^{2012}) = 3^{2012} + 2$$

9.(B) $f(x) = \tan^{-1} \left(\frac{2(\sqrt{3}-1)}{x^2 + \frac{3}{x^2} + 2} \right) x^2 + \frac{3}{x^2} \geq 2\sqrt{3} \therefore x^2 + \frac{3}{x^2} + 2$

$$\geq 2(\sqrt{3}+1)$$

$$\therefore f_{\max} = \tan^{-1} \left(\frac{2(\sqrt{3}-1)}{2(\sqrt{3}+1)} \right) = \frac{\pi}{12} = M \text{ which occurs at } x^2 = \frac{3}{x^2}$$

$$\text{i.e., at } x = 3^{1/4} = a$$

$$\text{and } f_{\min} = 0 = m \text{ at } x = 0$$

$$\cos^{-1} x + \cos^{-1} y = 3 \left(\tan^{-1} \left(\tan \frac{7\pi}{24} \right) + \tan^{-1} \left(\tan \frac{3\pi}{8} \right) \right)$$

$$= 3 \left(\frac{7\pi}{24} + \frac{3\pi}{8} \right) = 2\pi \Rightarrow x = y = -1 \Rightarrow x + y = -2$$

$$10.(D) \quad 3\sin^{-1}(\sin M) = 3\sin^{-1}\left(\sin \frac{\pi}{12}\right) = \frac{\pi}{4}$$

$$\therefore x^2 - \left(\tan\left(3\sin^{-1}(\sin M)\right)\right)x + a^4 = 0$$

$$\Rightarrow x^2 - (\tan \pi/4)x + 3 = 0; \quad x^2 - x + 3 = 0$$

$$\therefore \alpha + \beta = 1, \alpha\beta = 3 \quad \therefore (\alpha\beta) - (\alpha + \beta) = 3 - 1 = 2$$

11.(B) 12.(C)

Two points of inflection occurs at $x = 1$ and $x = 0$

$$\Rightarrow P''(1) = P''(0) = 0 \quad \Rightarrow P''(x) = ax(x-1)$$

$$\Rightarrow P'(x) = \frac{ax^3}{3} - \frac{ax^2}{2} + b \quad \therefore \left|\frac{dy}{dx}\right|_{x=0} = \tan(\sec^{-1}\sqrt{2}) = 1$$

$$P'(0) = 1 \Rightarrow b = 1$$

$$\therefore P'(x) = \frac{ax^3}{3} - \frac{ax^2}{2} + 1 \quad \therefore P(x) = \frac{ax^4}{12} - \frac{ax^3}{6} + x + c$$

$$P(-1) = 1 \therefore a\left(\frac{1}{12} + \frac{1}{6}\right) - 1 + c = 1$$

$$\Rightarrow \frac{a}{4} + c = 2 \quad \dots(1)$$

$$\text{and } P(1) = 1 \Rightarrow a = 12c \quad \dots(2)$$

solving (1) and (2), we get $c = \frac{1}{2}$ and $a = 6$

$$\therefore P(x) = \frac{x^4}{2} - x^3 + x + \frac{1}{2}; \quad P(0) = \frac{1}{2}$$

$$P'(x) = 0 \Rightarrow 2x^3 - 3x^2 + 1 = 0 \Rightarrow (2x+1)(x-1)^2 = 0$$

$$1.(4030) \quad \lim_{x \rightarrow 0} \sum_{k=1}^{2013} \frac{\left\{ \frac{x}{\tan x} + 2k \right\}}{2013} = \lim_{x \rightarrow 0} \left\{ \frac{x}{\tan x} \right\}$$

$$\lim_{x \rightarrow 0} \left(\frac{x}{\tan x} - \left[\frac{x}{\tan x} \right] \right) = \lim_{x \rightarrow 0} \left(\frac{x}{\tan x} \right) = 1 \quad \therefore 1^{\text{st}} \text{ term of G.P.} = 1$$

$$\text{Common ratio} = \lim_{x \rightarrow 0} \frac{e^{x^3} \left(e^{\tan^3 x - x^3} - 1 \right) (\tan^3 x - x^3)}{2 \ln(1 + x^3 \sin^2 x) \cdot x^3 \sin^2 x \cdot (\tan^3 x - x^3)}$$

$$= \frac{1}{2} \lim_{x \rightarrow 0} \left(\frac{\tan x - x}{x^3} \right) \left(\frac{\tan^2 x + x^2 + x \tan x}{x^2} \right) = \frac{1}{2} \cdot \frac{1}{3} \times 3 = \frac{1}{2}$$

$$\therefore S_{\infty} = \frac{1}{1 - \left(\frac{1}{2}\right)} = 2$$

2.(43) Since $f(x)$ has its domain as R

$$\therefore a - 3 = 0 \Rightarrow a = 3$$

$$f(x) = \frac{bx}{x^2 + 4} = y$$

$$bx = x^2y + 4y \Rightarrow x^2y - bx + 4y = 0$$

$\therefore x$ is real

$$\therefore D \geq 0$$

$$b^2 - 16y^2 \geq 0 \Rightarrow (4y - b)(4y + b) \leq 0$$

$$y \in \left[\frac{-b}{4}, \frac{b}{4} \right] \equiv [-8, 8]$$

$$\frac{b}{4} = 8 \Rightarrow b = 32$$

$$f(x) = \frac{32x}{x^2 + 4}$$

$$f'(x) = 32 \left(\frac{(x^2 + 4) \cdot 1 - x \cdot 2x}{(x^2 + 4)^2} \right) = \frac{32(4 - x^2)}{(x^2 + 4)^2}$$

$$f'(x) > 0 \Rightarrow x \in (-2, 2) \Rightarrow f(x) \uparrow$$

$$f'(x) < 0 \Rightarrow x \in (-\infty, -2) \cup (2, \infty) \Rightarrow f(x) \text{ is } \downarrow$$

Slope of the tangent at origin

$$\left. \frac{dy}{dx} \right|_{x=0} = 32 \times \left(\frac{4}{16} \right) = 8$$

Now, equation $f(x) = mx$ gives three solutions is $m \in (0, 8)$

$$\therefore a + b + p + q = 3 + 32 + 0 + 8 = 43$$

3.(0) $A = [a_{ij}]_{3 \times 3}$

$$a_{12} = 4, a_{13} = 5, a_{23} = 7$$

$$a_{11} = a_{22} = a_{33} = 0$$

$$a_{21} = -4, a_{31} = -5, a_{32} = -7$$

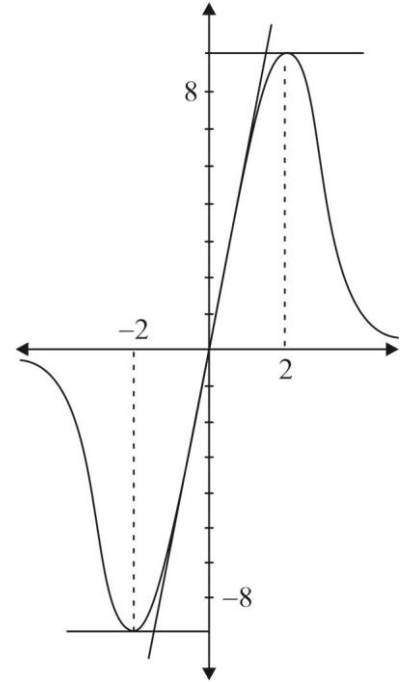
$$\therefore A = \begin{bmatrix} 0 & 4 & 5 \\ -4 & 0 & 7 \\ -5 & -7 & 0 \end{bmatrix} \Rightarrow |A| = 0$$

$$B = \begin{bmatrix} 0 & -4 & -5 \\ 4 & 0 & -7 \\ 5 & 7 & 0 \end{bmatrix} = -A$$

$$\therefore AB = A - (A) - 2A$$

$$|A - B| + |A - B|^2 + |A - B|^3 + \dots + |A - B|^{100}$$

$$|2A| + |2A|^2 + |2A|^3 + \dots + |2A|^{100} = 0$$



$$\begin{aligned}
 4.(2) \quad \therefore \quad \frac{\sin \alpha + \sin \beta + \sin \gamma}{\sin(\alpha + \beta + \gamma)} &= \frac{\cos \alpha + \cos \beta + \cos \gamma}{\cos(\alpha + \beta + \gamma)} \\
 &= \frac{\sin(\alpha + \beta + \gamma)(\sin \alpha + \sin \beta + \sin \gamma) + \cos(\alpha + \beta + \gamma)(\cos \alpha + \cos \beta + \cos \gamma)}{\sin(\alpha + \beta + \gamma)\sin(\alpha + \beta + \gamma) + \cos(\alpha + \beta + \gamma)\cos(\alpha + \beta + \gamma)} \\
 &\quad \left(\because \frac{a}{b} = \frac{c}{d} = \frac{xa + yc}{xb + yd} \right) \text{ (by ratio and proportion)} \\
 &= \frac{\cos(\alpha + \beta) + \cos(\alpha + \gamma) + \cos(\gamma + \alpha)}{1} = 2 \quad \Rightarrow \quad a = 2
 \end{aligned}$$

$$\therefore \quad \lim_{x \rightarrow 2} \frac{\sqrt{x^2 - 4}}{\sqrt{x - 2} + \sqrt{x} - \sqrt{2}} = 2$$

$$\begin{aligned}
 5.(0) \quad g'(x) &= 2x f' \left(\frac{x^2 + 2}{3} \right) - 2x f'(6 - x^2) = 2x \left(f' \left(\frac{x^2 + 2}{3} \right) - f'(6 - x^2) \right) \\
 f''(x) > 0 &\Rightarrow f' \text{ is increasing} \\
 f' \left(\frac{x^2 + 2}{3} \right) > f'(6 - x^2) &\Rightarrow \frac{x^2 + 2}{3} > 6 - x^2 \Rightarrow x^2 + 2 > 18 - 3x^2 \\
 4x^2 > 16 &\Rightarrow x^2 - 4 > 0 \Rightarrow x \in (-\infty, -2) \cup (2, \infty) \\
 g'(x) > 0 \quad \forall x \in (2, \infty) \cup (-2, 0) &\therefore a + b + c = 0
 \end{aligned}$$

$$6.(11) \quad \text{Let } A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \quad B = \begin{bmatrix} a_{12} + a_{13} & a_{11} + a_{13} & a_{11} + a_{12} \\ a_{22} + a_{23} & a_{21} + a_{23} & a_{21} + a_{22} \\ a_{32} + a_{33} & a_{31} + a_{33} & a_{31} + a_{32} \end{bmatrix}$$

$$\text{Now } |A| = 1$$

$$\begin{aligned}
 A^{-1}B &= \frac{1}{|A|} \begin{bmatrix} c_{11} & c_{21} & c_{31} \\ c_{12} & c_{22} & c_{32} \\ c_{13} & c_{23} & c_{33} \end{bmatrix} \begin{bmatrix} a_{12} + a_{13} & a_{11} + a_{13} & a_{11} + a_{12} \\ a_{22} + a_{23} & a_{21} + a_{23} & a_{21} + a_{22} \\ a_{32} + a_{33} & a_{31} + a_{33} & a_{31} + a_{32} \end{bmatrix} \\
 &= \frac{1}{|A|} \begin{bmatrix} 0 & |A| & |A| \\ |A| & 0 & |A| \\ |A| & |A| & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \Rightarrow n = |A^{-1}B| = 2
 \end{aligned}$$

$$\text{Now } A^{-1}B \begin{bmatrix} p \\ q \\ r \end{bmatrix} = A^{-1} \begin{bmatrix} 6 \\ 4 \\ 8 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ 4 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} p \\ q \\ r \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ 4 \end{bmatrix} \Rightarrow \begin{aligned} q + r &= 3 \\ p + r &= 2 \\ p + q &= 4 \end{aligned}$$

$$p = \frac{3}{2}, \quad q = \frac{5}{2}, \quad r = \frac{1}{2} \quad \therefore \quad 2(p + q + r) + n = 9 + 2 = 11$$